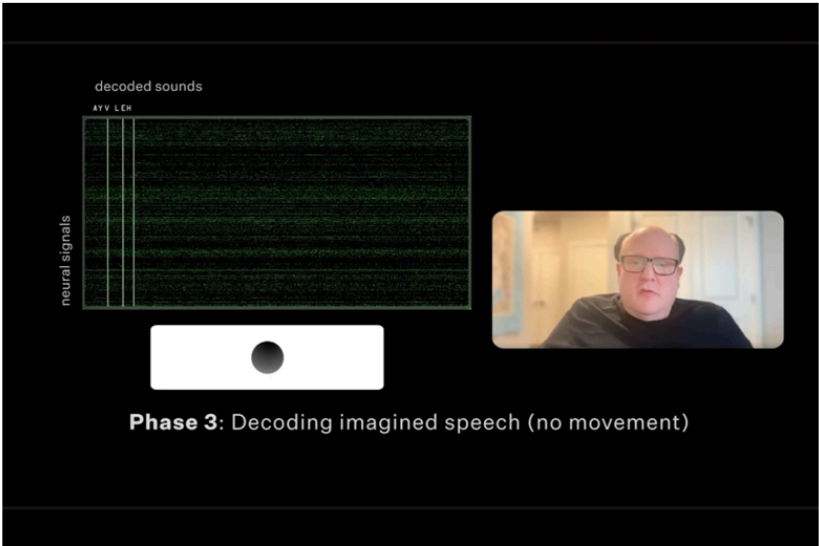


Restoring Speech to Paralyzed Patients: The History of Speech BCI

Can paralyzed patients speak without opening their mouths, just by thinking? This deck walks through the history of invasive speech brain-computer interfaces, from the discovery of high-gamma signals to the current stage of long-term implantation and imagined speech.

Neuralink – Imagined Speech

Bruce Yixuan Li



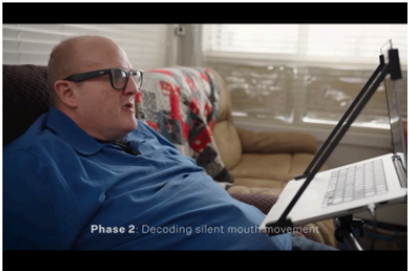
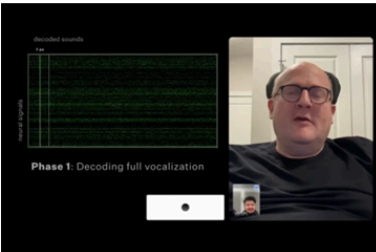
<https://www.youtube.com/@neuralink>

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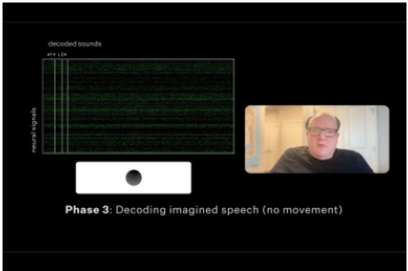
Hello, I'm Yixuan. Can paralyzed patients speak without opening their mouths, just by thinking? Are invasive speech BCIs really that powerful? On March 25, Neuralink released a new video showing exactly this process.

Read Aloud -> Mouth -> Imagined

Bruce Yixuan Li



- Which Signal?
- Which Model?
- How To Prove Safety?



<https://www.youtube.com/@neuralink>

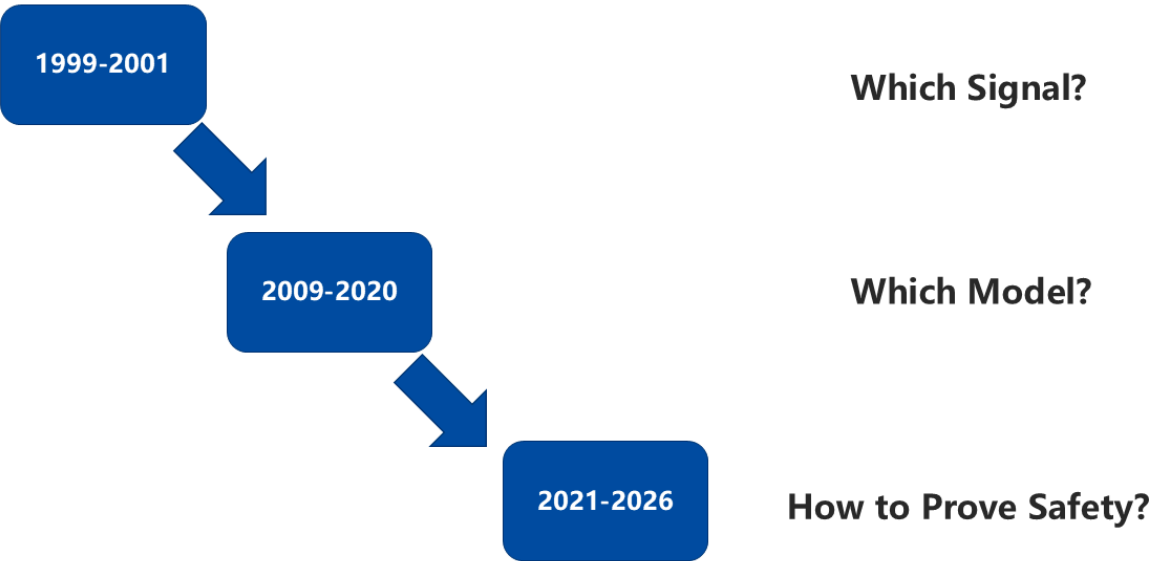
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Of course, this result was not achieved in a single step. The training had three stages: first, the patient read the text on the screen aloud. Second, the patient silently mouthed the text on the screen. Third, the patient simply "thought" the words in their mind. That is what we just saw.

The model for each stage had to be trained before moving to the next one. If you think carefully about this process, there are three problems to solve here. First, what signal should be decoded? Second, what model should be used for decoding? Third, how do we prove the safety of the BCI?

Three Stages

Bruce Yixuan Li



In fact, the development of speech BCIs has unfolded in exactly these three stages: first, figuring out what signal to decode; then, figuring out what model to use; and now, reaching the stage of verifying long-term implantation safety.

Next, I'll walk you through the history of speech BCIs. If you only care about recent progress, you can jump straight to Chapter 4.

Stage 1: 1999-2001, which signal to decode?

Which Signal?

Bruce Yixuan Li

Brain (1998), 121, 2301-2315

Functional mapping of human sensorimotor cortex with electrocorticographic spectral analysis
II. Event-related synchronization in the **gamma band**

Nathan E. Crone,¹ Diana L. Miglioretti,³ Barry Gordon^{1,4,5} and Ronald P. Lesser^{1,2,5}

Departments of ¹Neurology and ²Neurosurgery, The Johns Hopkins University School of Medicine, and the Departments of ³Biostatistics and ⁴Cognitive Science, and ⁵Zanvyl Krieger Mind/Brain Institute, The Johns Hopkins University, Baltimore, Maryland, USA

Correspondence to: Dr Nathan Crone, Department of Neurology, The Johns Hopkins Hospital, Meyer 2-147, 600 N. Wolfe St, Baltimore, MD 21287, USA.
E-mail: ncrone@jhmi.edu

Brazier Award-winning article, 2001

Induced electrocorticographic **gamma activity** during auditory perception

Nathan E. Crone^{a,*}, Dana Boatman^{a,b}, Barry Gordon^{a,c}, Lei Hao^a

^aDepartment of Neurology, The Johns Hopkins University School of Medicine, N. Wolfe St., Meyer Building, Baltimore, MD 21287-7247, USA

^bDepartment of Otolaryngology, The Johns Hopkins University School of Medicine, Baltimore, MD, USA

^cThe Zanvyl Krieger Mind/Brain Institute, The Johns Hopkins University, Baltimore, MD, USA

Accepted 30 November 2000

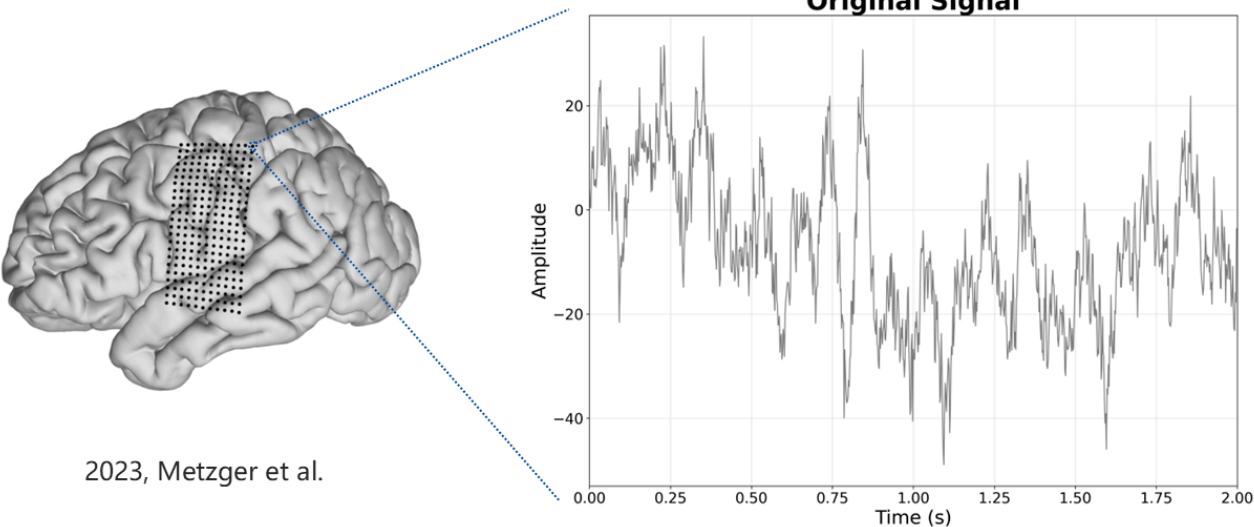
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Let's start with two papers from 1998 and 2001. At that time, a U.S. team unexpectedly discovered that high-gamma signals in the human brain were directly related to heard vowels and consonants. One of the researchers, Nathan Crone, later became a key figure in this field.

Honestly, this was absolutely one of the most important discoveries in the entire history of brain-computer interfaces.

Meaning of High Gamma ?

Bruce Yixuan Li



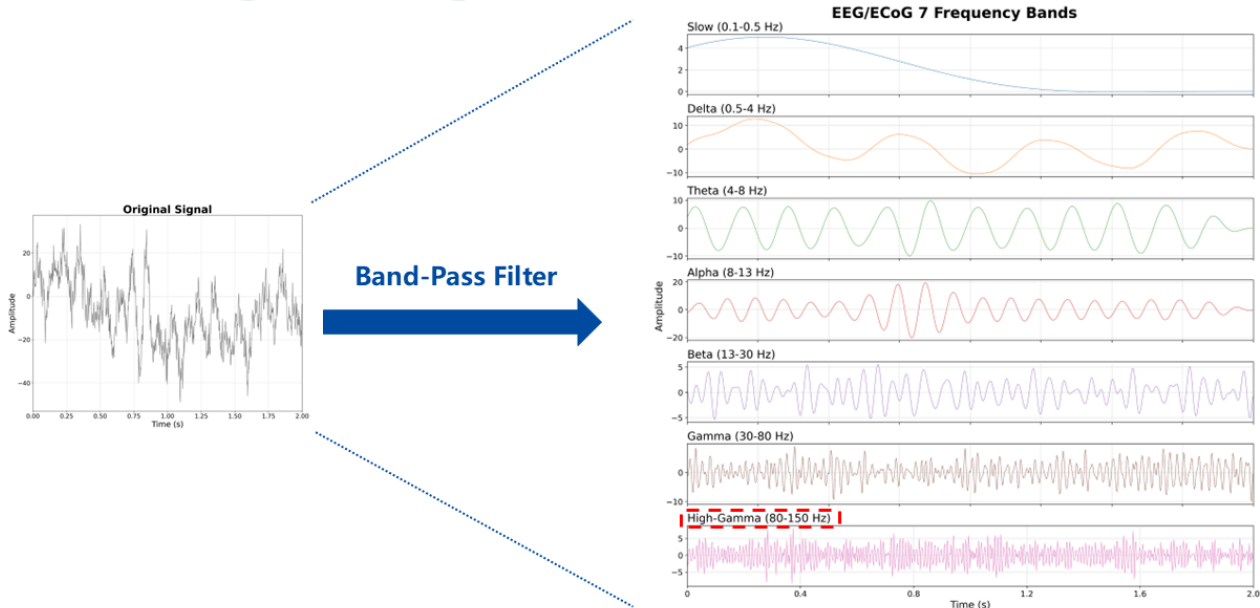
2023, Metzger et al.

2026/4/13	2026	1999-2001	2009-2020	2021-2026	2027-	Page: 5
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What does the high-gamma signal mean? The signal recorded at each electrode site is itself a mixture of periodic and aperiodic components.

Meaning of High Gamma ?

Bruce Yixuan Li



With a band-pass filter, we can separate it into different frequency bands. Researchers in the field conventionally divided it into seven bands. Any band above 80 Hz is what people in the field call the high-gamma signal.

EEG vs ECoG / sEEG

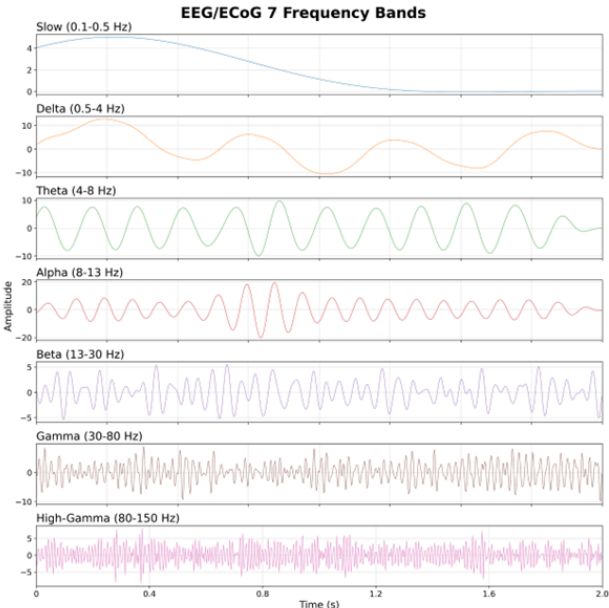
Bruce Yixuan Li

EEG



tissue	mean resistivity	lower and upper bound
brain white matter	700	350 and 1050 ($\pm 50\%$)
brain gray matter	300	150 and 450 ($\pm 50\%$)
spinal cord and cerebellum	650	325 and 975 ($\pm 50\%$)
cerebrospinal fluid	65	32.5 and 97.5 ($\pm 50\%$)
hard bone	16,000	8000 and 50,000
soft bone	2500	1250 and 3750 ($\pm 50\%$)
blood	160	80 and 240 ($\pm 50\%$)
muscle	1000	200 and 1800 ($\pm 80\%$)
fat	2500	1500 and 5000
eye	200	100 and 400
scalp	230	115 and 345 ($\pm 50\%$)
soft tissue	500	250 and 750 ($\pm 50\%$)
internal air	50,000	50,000 and 100,000

1997, Haueisen et al. **ECoG / sEEG**

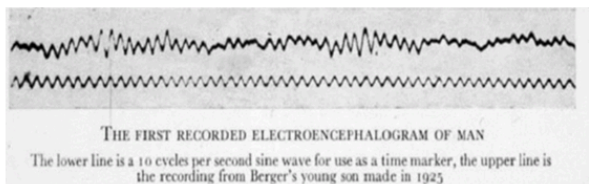


Scalp EEG, also called non-invasive EEG, can cover only 0.1 Hz to 80 Hz. Only intracranial EEG, also called invasive EEG, can capture signals above 80 Hz. This is because the human skull has very high electrical resistance, nearly the highest of any tissue in the body, so only low-frequency signals can pass through it.

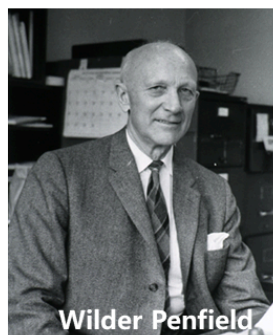
Why Delay

Bruce Yixuan Li

1924, EEG



1950s, ECoG & sEEG



Wilder Penfield



Jean Talairach

1973, BCI

TOWARD DIRECT BRAIN-COMPUTER COMMUNICATION

JACQUES J. VIDAL¹

Brain Research Institute,

University of California, Los Angeles, California

Why Was the Relationship Between High Gamma and Language Discovered So Late?
(1998 or 2001)

2026/4/13

2026

1999-2001

2009-2020

2021-2026

202?-

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That is why the relationship between high-gamma signals and language was discovered so late. Non-invasive EEG was invented in 1924; invasive EEG was developed in the 1950s by two surgeons; brain-computer interfaces were proposed in 1973. Yet the cornerstone of speech BCIs was not discovered until 1998.

Three Stages

Bruce Yixuan Li

1999-2001

Which Signal? --- High Gamma

2009-2020

Which Model?

2021-2026

How to Prove Safety?

2026/4/13

2026

1999-2001

2009-2020

2021-2026

202?-

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Now that humans know what signal to decode, the next step is to study what model to use for decoding.

Stage 2: 2009-2020, which model for decoding?

Which Model?

Bruce Yixuan Li



ML -----> DL
2019

2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 10
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Now the input is a high-gamma time series, and the output is text or speech. So how do we complete this process?

Because the input signal is produced by a complex system composed of many units, and machine learning and deep learning are especially good at handling such complex systems. Like many other fields, our field also experienced a shift from machine learning to deep learning. In our field, that shift happened around 2019.

From ML to DL

Bruce Yixuan Li

A Wireless Brain-Machine Interface for Real-Time Speech Synthesis Frank H. Guenther^{1,2*}, Jonathan S. Brumberg^{1,3}, E. Joseph Wright³, Alfonso Nieto-Castanon⁴, Jason A. Tourville³, Mikhail Panko¹, Robert Law¹, Steven A. Siebert³, Jess L. Bartels³, Dinal S. Andreasen^{3,5}, Princewill Ehirim⁶, Hui Mao⁷, Philip R. Kennedy³	Brain-computer interfaces for speech communication Jonathan S. Brumberg^{a,c,*}, Alfonso Nieto-Castanon^f, Philip R. Kennedy^e, Frank H. Guenther^{a,b,c,d}	Handcraft Feauters & SVM				
Direct classification of all American English phonemes using signals from functional speech motor cortex Emily M. Mugler¹, James L. Patton¹, Robert D. Flint¹, Zachary A. Wright¹, Stephan U. Schuele¹, Joshua Rosenow¹, Jerry J. Shih¹, Dean J. Krusienski¹ and Marc W. Slutzky^{1,2,3,4}	Brain-to-text: decoding spoken phrases from phone representations in the brain Christian Herff^{1,†}, Dominic Heger^{1,†}, Adriana de Pesters^{2,3}, Dominic Telaar¹, Peter Brunner^{2,4}, Gerwin Schalk^{2,3,4} and Tanja Schultz¹	ML Brain->Phonemes				
2019						
Speech synthesis from neural decoding of spoken sentences Gopala K. Anumanchipalli^{1,2,4}, Josh Chartier^{1,2,3,4} & Edward F. Chang^{1,2,3,*}	Speech synthesis from ECoG using densely connected 3D convolutional neural networks Miguel Angrik^{1,2}, Christian Herff^{1,2,3}, Emily Mugler¹, Matthew C Tate¹, Marc W Slutzky^{1,2}, Dean J Krusienski¹ and Tanja Schultz¹	Brain->Sentences				
Brain2Char: a deep architecture for decoding text from brain recordings Pengfei Sun^{1,2}, Gopala K Anumanchipalli^{1,2}, Edward F Chang^{1,3}	Machine translation of cortical activity to text with an encoder-decoder framework Joseph G. Makin^{1,2,3}, David A. Moses^{1,2} and Edward F. Chang^{1,2,3}	DL LSTM				
2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 11

Before 2019, papers all relied on manually extracting features and then adding an SVM classifier. They could only decode phonemes. After 2019, sentence decoding became possible. The most effective model was the LSTM.

Many people ask: why didn't anyone try deep learning before 2019? You have to remember that AlexNet in 2012 was a hand-built CNN, and people even had to manually coordinate communication between two GPUs. Anyone without a computer science background simply couldn't manage that. It was not until TensorFlow in 2015 and PyTorch in 2016 that people no longer had to hand-build models. With about ten lines of code, you could set up an RNN, and the barrier to entry dropped dramatically.

Short Term Implantation

Bruce Yixuan Li

Brain Tumor (≈30 min)

Speech synthesis from neural decoding of spoken sentences

Gopala K. Anumanchipalli^{1,2,4}, Josh Chartier^{1,2,3,4} & Edward F. Chang^{1,2,3,4}

Decoding and synthesizing tonal language speech from brain activity

Yan Liu^{1,2,3,4}, Zehao Zhao^{1,2,3,4}, Minpeng Xu^{5,6}, Haiping Yu⁵, Yanming Zhu^{1,2,3,4}, Jie Zhang^{1,2,3,4}, Linghao Bu^{1,2,3,4,7}, Xiaolu Zhang^{1,2,3,4}, Junfeng Lu^{1,2,3,4,8,9}, Yuanming Li^{1,2}, Dong Ming^{5,6}, Jinsong Wu^{1,2,3,4}

A brain-to-text framework for decoding natural tonal sentences

Daohan Zhang, Zhenjie Wang, Youkun Qian, ..., Kunyu Xu, Yuanming Li, Junfeng Lu

Epilepsy (7~14 days)

Brain-to-text: decoding spoken phrases from phone representations in the brain

Christian Herff^{1**}, Dominic Heger^{1**}, Adriana de Pesters^{2,3}, Dominic Telaar¹, Peter Brunner^{2,4}, Gerwin Schaik^{2,3,4} and Tanja Schultz¹

Direct classification of all American English phonemes using signals from functional speech motor cortex

Emily M Mugler¹, James L Patton¹, Robert D Fildes¹, Zachary A Wright¹, Stephan M Schuele², Joshua Roseboom², Jerry J Shih², Dean J Krusienski¹ and Marc W Slutzky^{1,5,6}

Speech synthesis from ECoG using densely connected 3D convolutional neural networks

Miguel Angrick^{1,2}, Christian Herff^{1,2}, Emily Mugler¹, Matthew C Tate¹, Marc W Slutzky^{1,2}, Dean J Krusienski¹ and Tanja Schultz¹

Brain2Char: a deep architecture for decoding text from brain recordings

Pengfei Sun^{1,2}, Gopala K Anumanchipalli^{1,2}, Edward F Chang^{1,3}

Machine translation of cortical activity to text with an encoder-decoder framework

Joseph G. Makin^{1,2,10}, David A. Moses^{1,2} and Edward F. Chang^{1,2,10}

Patients Who Really Need Speech BCI: ALS, Stroke.

2026/4/13

2026

1999-2001

2009-2020

2021-2026

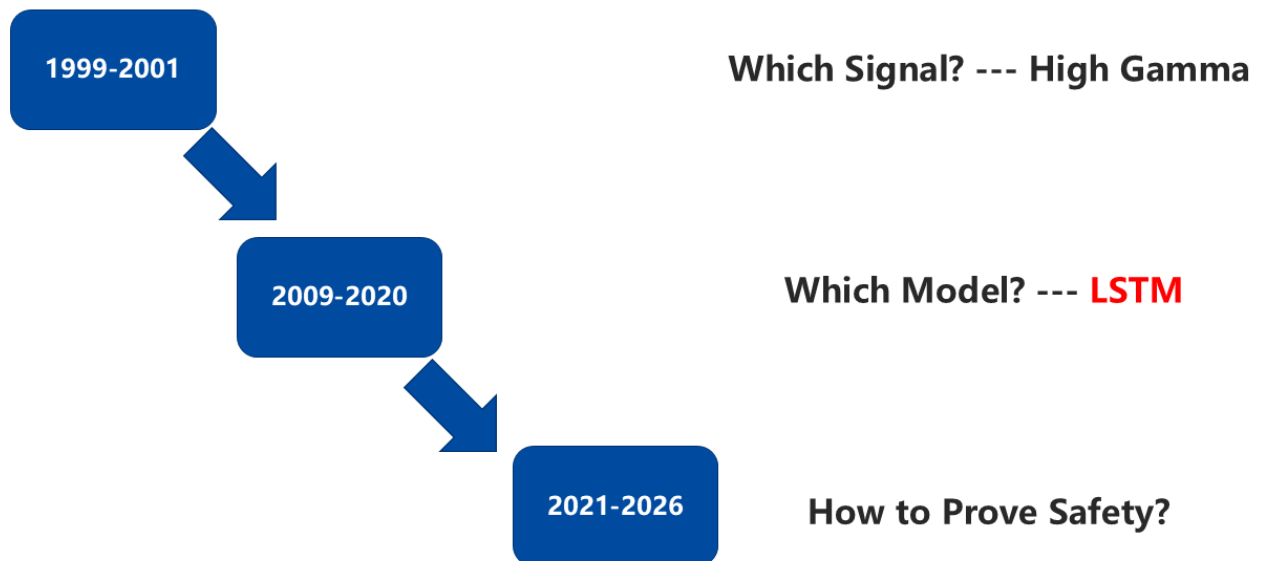
2027-

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It should be pointed out that all of these studies were done in patients with short-term implants, usually people with brain tumors or epilepsy. But the patients who truly need speech BCIs are those with ALS or stroke.

Three Stages

Bruce Yixuan Li



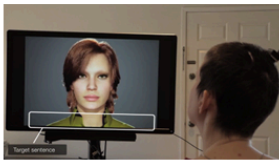
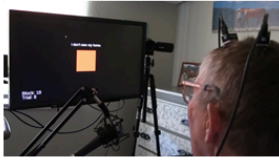
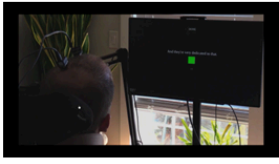
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Studies with short-term implants can tell us that LSTMs are effective, but they cannot tell us whether long-term implanted speech BCIs are safe.

Stage 3: 2021-2026, Safety of long term implantation

Long Term Implantation

Bruce Yixuan Li

	Brain -> Text (-> Audio)			Brain -> Audio
	 2021, UCSF Corpus: 50 WER: 25%	 2023, UCSF Corpus: 1024 WER: 25.5%	 2024, UCSF Corpus: 104 WER: 25%	 2025, UCSF Corpus: 1024 WER: 58.8%
LSTM/GRU & N-Gram LM		 2023, BrainGate Corpus: 125K WER: 23.8%	 2024, BrainGate Corpus: 125K WER: 2.5%	 2025, BrainGate Corpus: Infinite WER: 43.75%
LSTM/GRU & N-Gram LM & GPT-Style LM				

2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 14
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Finally, in 2021, the University of California, San Francisco (UCSF) produced the world's first long-term implanted speech BCI that could decode sentences. This patient had muscular atrophy, and they asked the patient to read the text on the screen aloud. They achieved a 25% word error rate with a 50-word vocabulary.

Here's a small tip: when judging whether a BCI paper is reliable, see whether the authors dare to release complete supplementary videos. That one trick filters out most of the fluff.

In 2023, UCSF added a digital avatar on top of this and expanded the vocabulary from 50 to 1,024, with almost no change in word error rate.

In the same year, on another electrode material, the Utah Array, the BrainGate team expanded the vocabulary from 50 to 125,000. Keep in mind that everyday English uses only about 10,000 common words, so 125,000 already includes many names and places. Their word error rate was 23.8%.

In 2024, the UCSF team built a bilingual BCI with a vocabulary of 104 and a word error rate of 25%.

In the same year, the BrainGate team made an even bigger breakthrough: on a 125,000-word vocabulary, they reduced the word error rate to 2.5%. A key reason was that their participant was in the early stage of ALS and in better condition.

In 2025, the UCSF team achieved brain-to-voice, allowing the patient to control pitch, loudness, and speaking rhythm, but the word error rate rose to 50%.

In the same year, BrainGate also achieved brain-to-voice.

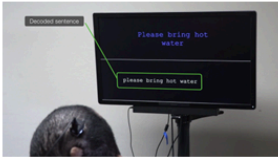
Listening to all this, you may feel that it's always just these two teams. And yes, before Neuralink joined, the only groups doing long-term implanted speech BCIs well were UCSF and BrainGate.

Long Term Implantation

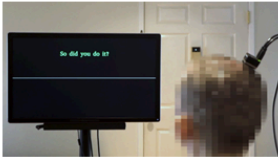
Attempt or Mouth



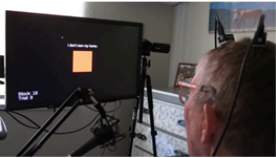
2023, UCSF
Corpus: 1024
WER: 25.5%



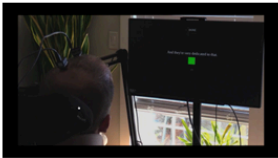
2024, UCSF
Corpus: 104
WER: 25%



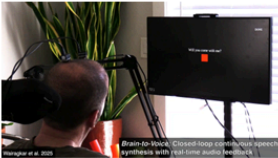
2025, UCSF
Corpus: 1024
WER: 58.8%



2023, BrainGate
Corpus: 125K
WER: 23.8%

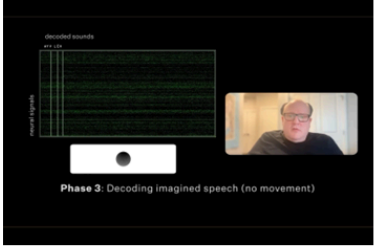


2024, BrainGate
Corpus: 125K
WER: 2.5%



2025, BrainGate
Corpus: Infinite
WER: 43.75%

Imagine



2026, Neuralink
Corpus: ?
WER: ?

2026/4/13

2026

1999-2001

2009-2020

2021-2026

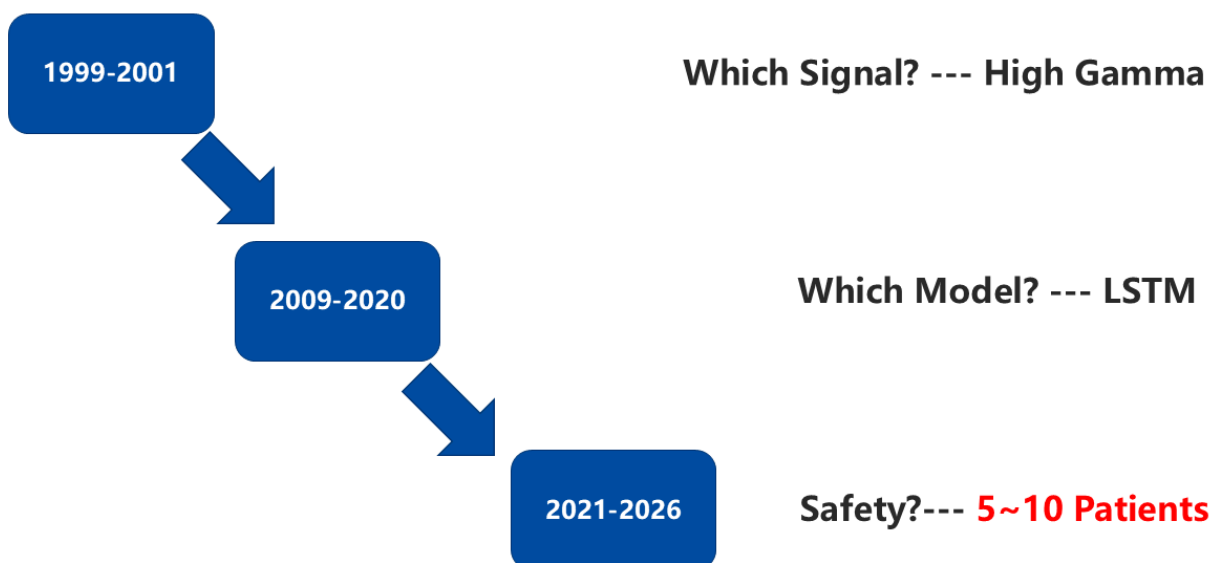
202?-

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In 2026, Neuralink released a video, but they did not publish their vocabulary size or word error rate. Still, one major advance is clear: in the earlier papers, participants either read aloud or silently mouthed the words, whereas Neuralink reached pure imagined speech.

Three Stages

Bruce Yixuan Li

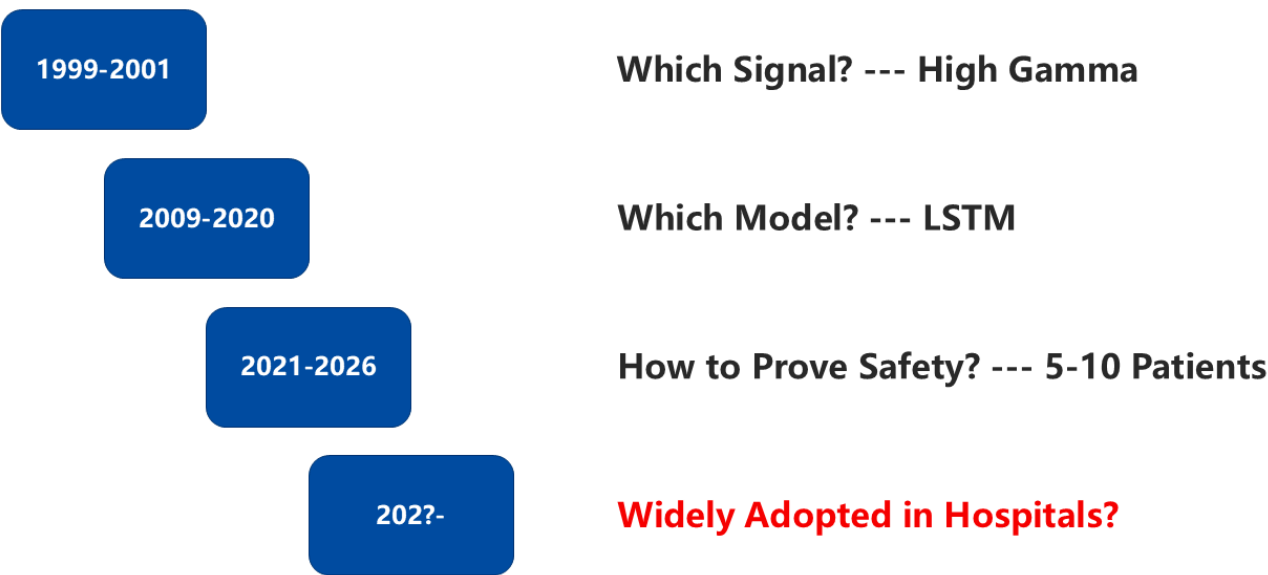


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There are now 5 to 10 patients with long-term implants, and their very existence already demonstrates the long-term safety of speech BCIs. The reason is simple: if the devices were not safe, these patients would not have made it this far.

The 4th Stage

Bruce Yixuan Li



2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 17
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But at present, speech BCIs still have not entered major hospitals the way coronary bypass surgery has.

1. What If The BCI Performance Go Down After 2 Years?
2. How to Solve High Variability Among Patients?



We Need A Fallback Algorithm.

2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 18
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In my view, there are two key problems that must be solved to get there in the future. First, after many years of implantation, when BCI performance declines, how can patients continue to benefit? Second, patients differ in disease severity, so the help provided by a BCI also differs. How can we ensure that implantation makes life easier for every patient, not just some of them?

In essence, both problems come down to designing a fallback algorithm. The ceiling is already very high: earlier work has enabled communication close to normal conversation. But the floor of speech BCIs has not yet been built.

We Need A Fallback Algorithm.

1. Open Corpus -> Closed Corpus
2. Brain2Voice -> Brain2Text + TTS
3. Imagine -> Attempt

2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 19
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I see three ways to do this. First, give patients 1,000 English words or 1,000 Chinese characters, enough to handle communication about eating, drinking, toileting, and other daily needs. Second, there is no need to pursue flashy brain-to-audio systems; brain-to-text is enough, and TTS can handle the rest afterward. Third, we also do not need to rely on imagined speech. It is enough if the patient simply tries hard to speak.

Q & A

Bruce Yixuan Li

1. Would Speech BCI Reveal Thoughts that the Patients Does Not Want to Express?

2. Are There Any Public Datasets Available for Others to Validate?

New Videos on This Channel Will Address These Two Questions, as Well as Yours.

Any Questions Are Welcomed. Please Leave Them in the Comments.

2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 20
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After watching this, you may have two questions. Could this method expose thoughts the patient does not want to say out loud? And are there any public datasets that people can use for validation? A new video will address these two questions, and you are also welcome to leave your own questions in the comments.

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Bruce Yixuan Li

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2026/4/13	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 21
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This is all for today.

Chinese

https://github.com/Physics-Lee/20260315_Speech_Neuroprosthesis/blob/master/20260315_History_of_Speech_BCI/CN.pdf

English

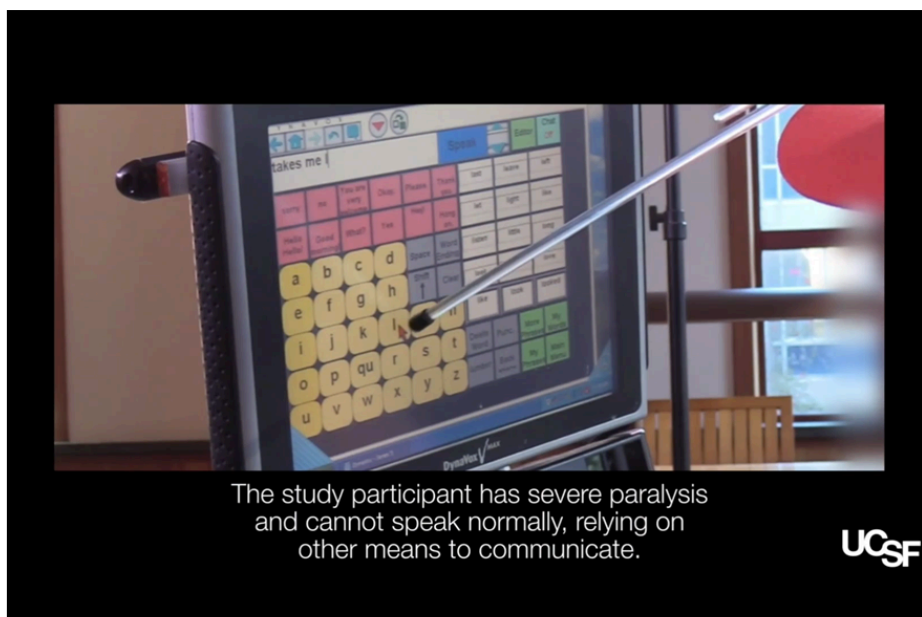
https://github.com/Physics-Lee/20260315_Speech_Neuroprosthesis/blob/master/20260315_History_of_Speech_BCI/EN.pdf

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You can find the transcripts of the video here.

2021, UCSF

Bruce Yixuan Li



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That concludes today's talk. The final minute is a 2021 UCSF promotional video. This was the world's first successful long-term implanted speech BCI to decode sentences.